HARD GEOMETRY AND TRIGONOMETRY

Circles

02

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at -4, -6. Line k is tangent to this circle at the point -7, -7. What is the slope of line k?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

Q2 GEOMETRY AND TRIGONOMETRY

$$x^2 + 14x + y^2 = 6y + 109$$

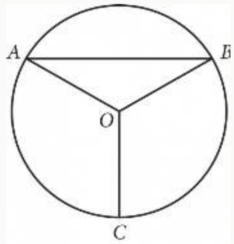
In the xy-plane, the graph of the given equation is a circle. What is the length of the circle's radius?

A. $\sqrt{109}$

B. $\sqrt{149}$

C. $\sqrt{167}$

D. $\sqrt{341}$



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30°. If the length of $\overline{OC}$ is 18, what is the length of arc $\overset{\frown}{AB}$?

A. 9π

B. 12π

C. 15π

D. 18π

Q4 GRID-IN GEOMETRY AND TRIGONOMETRY

Circle A in the xy-plane has the equation $x + 5^2 + y - 5^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy-plane is $x + 5^2 + y - 5^2 = k$, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN GEOMETRY AND TRIGONOMETRY

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at $-1, 1$. Line t is tangent to this circle at the point $5, -4$. Which of the following points also lies on line t?

A. $0, \frac{6}{5}$

B. $4, 7$

C. $10, 2$

D. $11, 1$

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at $-5, 2$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a, b, and c are constants. What is the value of c?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 GEOMETRY AND TRIGONOMETRY

$$(x-6)^2+(y+5)^2=16$$

In the xy-plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10,-5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

A. $(2,-5)$

B. $(6,-1)$

C. $(6,-5)$

D. $(6,-9)$

Q9 GEOMETRY AND TRIGONOMETRY

The equation $x^2 + y - 1^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the *xy*-plane. Which of the following equations represents circle B?

A. $x - 2^2 + y - 1^2 = 49$

B. $x^2 + y - 3^2 = 49$

C. $x + 2^2 + y - 1^2 = 49$

D. $x^2 + y + 1^2 = 49$

Q10

GEOMETRY AND TRIGONOMETRY

Points Q and R lie on a circle with center P. The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

A. $13\sqrt{2}$

B. 13

C. $9\sqrt{2}$

D. 9